Unit -1

Basics of pattern recognition, data sets for Pattern Recognition, Different Paradigms of Pattern Recognition, Design principles of pattern recognition system, Representations of Patterns and Classes, Learning and adaptation, Pattern recognition approaches, Metric and non-metric proximity measures, Mathematical foundations – Linear algebra, Probability Theory, Expectation, mean and covariance, Normal distribution, multivariate normal densities, Chi squared test.

Metric and non-metric proximity measures

Distance measure

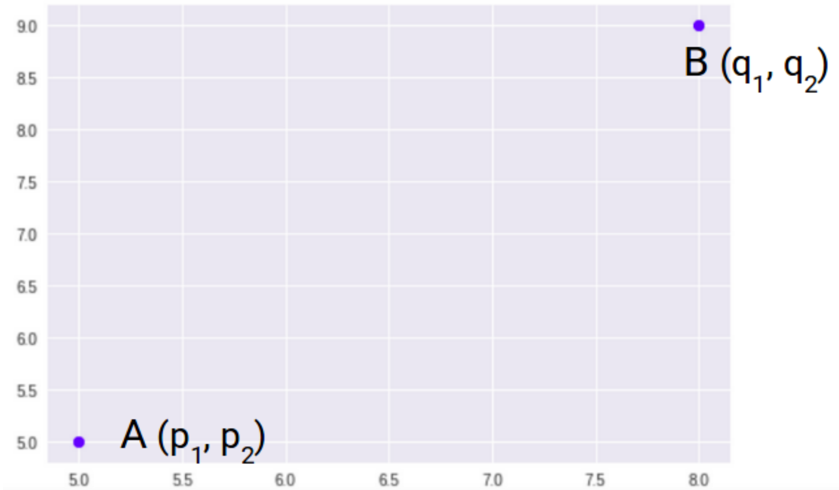
• These measures find the distance between points in a d-dimensional space, where each pattern is represented as a point in the d-space.

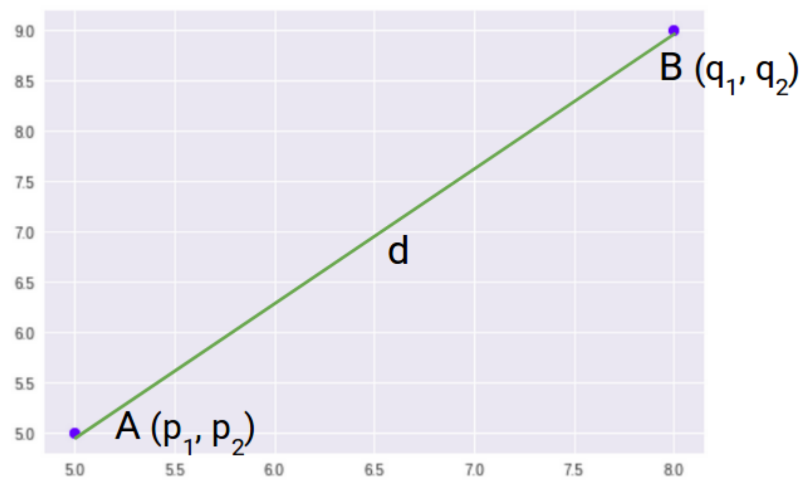
• The distance is inversely proportional to the similarity. If d(X,Y) gives the distance between X and Y , and s(X,Y) gives the similarity between X and Y , then $d(X, Y) \propto 1/ s(X,Y)$

Euclidean distance(ED)

The smallest distance between two points is known as the Euclidean Distance.

This distance metric is used by most machine learning algorithms, including K-Means, to assess the similarity of observations. Assume we have the following two points:





Here's the formula for Euclidean Distance:

$$d = ((p_1 - q_1)^2 + (p_2 - q_2)^2)^{1/2}$$

We use this formula when we are dealing with 2 dimensions. We can generalize this for an n-dimensional space as:

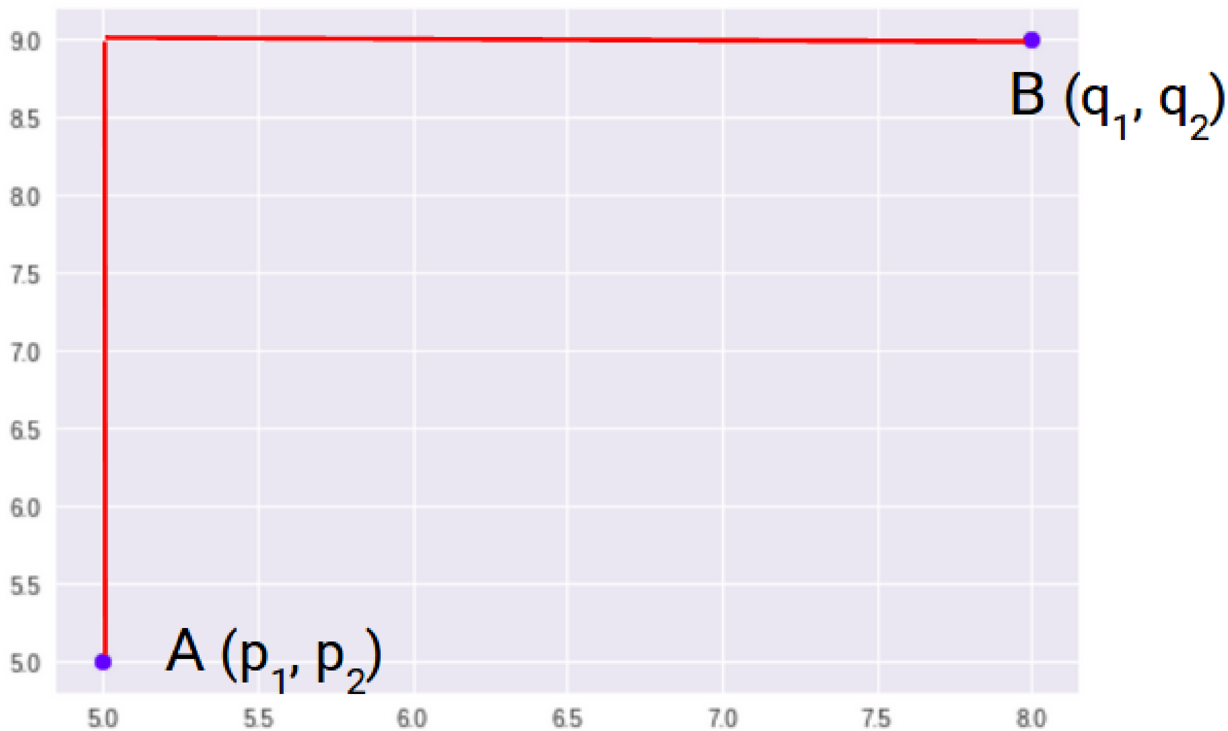
$$D_e = \left(\sum_{i=1}^{n} (p_i - q_i)^2 \right)^{1/2}$$

Where,

n = number of dimensions
pi, qi = data points

Manhattan distance:

Manhattan Distance is the sum of absolute differences between points across all the dimensions.



Since the above representation is 2 dimensional, to calculate Manhattan Distance, we will take the sum of absolute distances in both the x and y directions. So, the Manhattan distance in a 2-dimensional space is given as:

$$d = |p_1 - q_1| + |p_2 - q_2|$$

And the generalized formula for an n-dimensional space is given as:

$$D_m = \sum_{i=1}^{n} |p_i - q_i|$$

Where,

n = number of dimensions
pi, qi = data points

Minkowski Distance

Minkowski Distance is the generalized form of Euclidean and Manhattan Distance. The formula for Minkowski Distance is given as:

$$D = \left(\sum_{i=1}^{n} |p_i - q_i|^p \right)^{1/p}$$

Here, p represents the order of the norm.

Weighted distance measure
When some features are more significant than others, the weighted distance measure is used. The weighted distance is as follows :

$$d(X,Y) = (\sum_{k=1}^{d} w_k(x_k - y_k)^m)^{\frac{1}{m}}$$

- The weight for each feature depends on its importance.
- The greater the significance of the feature, the larger the weight.
- The weighted distance measure using Euclidean distance(m=2) would be

k-Median Distance

• The k-Median distance is a non-metric distance function.
• This finds the distance between two vectors.
• The difference between the values for each element of the vector is found.
• Putting this together, we get the difference vector.
• If the values in the difference vector is put in non-decreasing order, the k^{th} value gives the k Median distance.